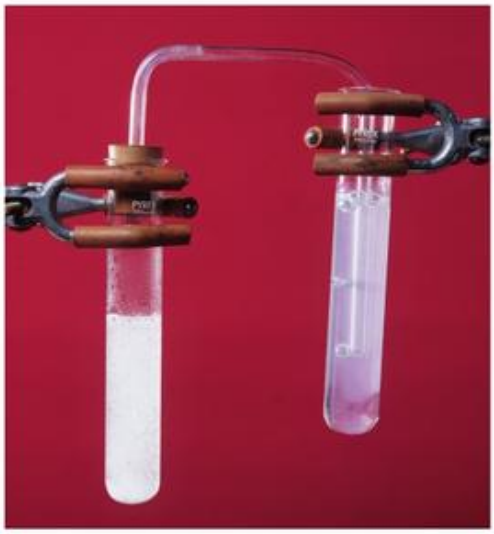
HINT

You can use any acid to do this test, but you have to be careful, especially with sulfuric acid, as some acid–carbonate combinations can produce an insoluble salt that coats the solid carbonate and stops the reaction. The test works well with dilute hydrochloric acid or dilute nitric acid.

For example, using zinc carbonate and dilute nitric acid:

$$ZnCO_3(s) + 2HNO_3(aq) \rightarrow Zn(NO_3)_2(aq) + CO_2(g) + H_2O(l)$$



▲ Figure 18.12 We can test to see if carbon dioxide is produced by bubbling the gas through limewater (in the tube on the right) and looking to see whether the limewater goes cloudy/milky (white precipitate of calcium carbonate formed).

! Safety Note: Wear eye protection for all these tests and avoid skin contact with the reactants and products.

TESTING FOR SULFATES (SO_4^{2-})

KEY POINT

You could also use nitric acid and barium nitrate solution in the same way.

Make a solution of your suspected sulfate in distilled (pure) water, add some *dilute hydrochloric acid* and then add some *barium chloride solution*. A sulfate will produce a white precipitate of barium sulfate.

$$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$$

You add acid to the solution to react with and remove other anions which might also produce white precipitates when you add the barium chloride solution. For example, if you didn't add acid, you would also obtain a white precipitate if there were carbonate ions present. This is because barium carbonate is also white and insoluble. The acid reacts with the carbonate ions and removes them. It is important to remember that the acid must be added *before* the barium chloride solution.

You must never acidify the solution with sulfuric acid because sulfuric acid contains sulfate ions. If you add those, you are bound to get a precipitate of barium sulfate because you have added sulfate ions.

TESTING FOR CHLORIDES (Cl^-), BROMIDES (Br^-) AND IODIDES (I^-)

! Safety Note: Wear eye protection for all these tests and avoid skin contact with the reactants and products.

KEY POINT

All of these precipitates tend to discolour to greys and pale purples on exposure to light.

Make a solution of your suspected chloride, bromide or iodide and add enough *dilute nitric acid* to make it acidic. Then add some *silver nitrate solution*.

A *white* precipitate (of silver chloride) shows the presence of chloride ions:

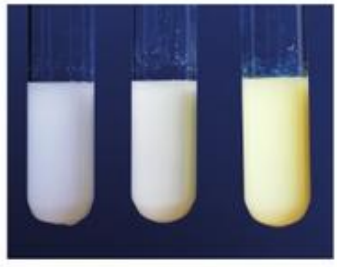
$$Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$$

A *cream* precipitate (of silver bromide) shows the presence of bromide ions:

$$Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$$

A *yellow* precipitate (of silver iodide) shows the presence of iodide ions:

$$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$$



▲ Figure 18.13 Precipitates of silver chloride, silver bromide and silver iodide

KEY POINT

The 1– ions formed by the halogens (Group 7) are collectively called *halide ions*. Chloride, bromide and iodide ions are all halide ions.

The acid is added to react with and remove other anions which might also produce precipitates with silver nitrate solution, for example carbonate and hydroxide ions. You must not use hydrochloric acid (or hydrobromic acid or hydroiodic acid) as you will be adding halide ions and you are certain to obtain a precipitate.

Table 18.2 A summary of the tests for negative ions (anions)

Ion	Test	Positive result	Product of reaction
CO_3^{2-}	add dilute hydrochloric acid then bubble any gas produced through limewater	fizzing, limewater goes cloudy (white precipitate)	CO_2 $CaCO_3$ (with limewater)
SO_4^{2-}	add dilute hydrochloric acid followed by barium chloride solution	white precipitate	$BaSO_4$
Cl^-	add dilute nitric acid followed by silver nitrate solution	white precipitate	AgCl
Br^-	add dilute nitric acid followed by silver nitrate solution	cream precipitate	AgBr
I^-	add dilute nitric acid followed by silver nitrate solution	yellow precipitate	AgI

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING

1 Name the gas being described in each of the following cases.

a A green gas that bleaches damp litmus paper.

b A gas that dissolves readily in water to produce a solution with a pH of about 11.

c A gas that produces a white precipitate with limewater.

d A gas that pops when a lighted splint is placed in it.

e A gas that relights a glowing splint.

SKILLS EXECUTIVE FUNCTION

2 Describe in detail how you would carry out the following tests. In each case state whether you would carry out the test on the solid or a solution, and describe what you would expect to see happen.

a A flame test for lithium ions in lithium chloride.

b A test for ammonium ions in ammonium sulfate.

c A test for sulfate ions in ammonium sulfate.

d A test for carbonate ions in calcium carbonate.

e A test for iodide ions in potassium iodide.

3 **A** is an orange solid, which dissolves in water to give an orange solution. When sodium hydroxide solution is added to a solution of **A**, an orange-brown precipitate, **B**, is formed. Adding dilute nitric acid and silver nitrate solution to a solution of **A** gives a white precipitate, **C**.

SKILLS CRITICAL THINKING

a State the names of **A**, **B** and **C**.

SKILLS PROBLEM SOLVING

b Write equations (full or ionic) for the reactions producing **B** and **C**.

SKILLS PROBLEM SOLVING, CRITICAL THINKING (6)

4 **D** is a green crystalline solid that dissolves in water to give a very pale green solution. Addition of sodium hydroxide solution to a solution of **D** produces a green precipitate, **E**, which turns orange-brown around the top after standing in air. Addition of dilute hydrochloric acid and barium chloride solution to a solution of **D** gives a white precipitate, **F**.

a State the names of **D**, **E** and **F**.

SKILLS PROBLEM SOLVING (9)

b Write equations (full or ionic) for the reactions producing **E** and **F**.

(6)

5 **G** is a colourless crystalline solid which reacts with dilute nitric acid to give a colourless solution, **H**, and a colourless, odourless gas, **I**, which turns limewater milky. **G** has a lilac flame colour.

SKILLS CRITICAL THINKING

a State the names of **G**, **H** and **I**.

SKILLS PROBLEM SOLVING (9)

b Write an equation (full or ionic) for the reaction between **G** and dilute nitric acid.

(6)

6 **J** is a colourless solid. When it is warmed with sodium hydroxide solution it produces a gas, **K**, which turns damp universal indicator paper blue. When nitric acid, followed by silver nitrate solution, is added to a solution of **J**, a cream precipitate of **L** is formed.

SKILLS CRITICAL THINKING

a State the names of **J**, **K** and **L**.

SKILLS PROBLEM SOLVING (9)

b Write an ionic equation for the formation of the cream precipitate.

SKILLS EXECUTVE FUNCTION (5)

7 A student has found that her sample of potassium nitrate is contaminated with small amounts of a green solid. She picks out a small piece of the green solid and finds that it is insoluble in water.

SKILLS CRITICAL THINKING (6)

a Describe how you would make a pure sample of potassium nitrate from the impure mixture.

b The student believes that the green solid could be copper(II) carbonate. Describe a series of tests that the student could use to confirm this.

(5)

8 A student finds a bottle containing a colourless liquid in the laboratory.

a Describe a test to show that the bottle contains water.

(4)

b Describe a test to show whether it is pure water.

SKILLS EXECUTIVE FUNCTION, REASONING (7)

c The student carried out the test in **b** and it showed that the water was not pure. He then carried out some more tests to try to identify the solution. The following results were obtained:

Test	Result
add sodium hydroxide solution	no visible reaction but the test-tube became warm
add dilute hydrochloric acid	no reaction
add dilute hydrochloric acid followed by barium chloride solution	no reaction
add nitric acid followed by silver nitrate solution	white precipitate

Explain what could be present in the bottle and how the student could confirm this.

UNIT QUESTIONS

You may need to refer to the Periodic Table on page 320.

SKILLS REASONING 8

1

Lithium, sodium and potassium are three elements in Group 1 of the Periodic Table. All these elements are stored under oil so that they cannot react with air and water.

a Explain in terms of the structure of the atoms why these three elements are all placed in Group 1 of the Periodic Table. **(1)**

SKILLS CRITICAL THINKING 5

b State the name of the product obtained when potassium reacts with air. **(1)**

c Only three of the statements about the reaction of lithium with water given below are true. Put a cross next to each of the correct statements. **(3)**

The water turns blue	☐
The piece of lithium floats	☐
The lithium burns with a lilac flame	☐
Hydrogen gas is formed	☐
The lithium explodes	☐
The final solution is alkaline	☐
Lithium reacts more violently than sodium	☐

d Some data about the physical properties of lithium, sodium and potassium are shown in the table.

	Melting point/°C	Boiling point/°C	Density/g/cm^3
Li	181	1342	0.53
Na	98	883	0.97
K	63	760	0.86

Caesium is another element in Group 1 of the Periodic Table.

7

i Which of the following are likely to be properties of caesium? Put crosses in two boxes. **(2)**

It forms a 1+ ion in compounds	☐
It has a melting point of 60 °C	☐
It is a non-metal	☐
It reacts with water and air	☐
It is an alkali	☐

SKILLS REASONING 8

ii Explain why the data in the table cannot be used to predict whether caesium has a higher or lower density than potassium. **(2)**

(Total 9 marks)